Statistical Analysis based on the first SAS program for Syfiber data

The SAS file 1 for Syfiber S2025 is our first SAS program for this data set. Note that every statement in the SAS program ends in a semi colon: ";" This is because SAS code is written in the PL1 programming language. The first SAS statement is TITLE and the name for it appears between single quotes. This statement is optional.

The next SAS statement is DATA. This has to be included in a SAS program and it gives a name to the analyzed data set. The next SAS statement is the INPUT statement. In this statement we list the variables in our data set. Note that before the semicolon we have the @@ symbols. This instructs the SAS system that we will have a free format data input. The following two variables are listed: PERCENT STRENGTH. Note that one has to have at least one space between the variable names. The data is listed after CARDS; statement. The @@ format implies that the first data value is assigned to PERCENT, the second to STRENGTH, the third to PERCENT, the fourth to strength, and so on until the end of the data set.

The LABEL statement, after the INPUT statement, is optional. It gives a more explicit name to the PERCENT variable.

The next three SAS statements generate most of the output for the statistical analysis of the one way ANOVA model. The PROC GLM PLOTS=(DIAGNOSTICS); statement generates important FIT Diagnostics plots as well as test of hypotheses for testing the null hypothesis that the treatment means are equal to each other, in a fixed effects model. The CLASS statement has to be included after the PROC GLM statement, as it specifies the variable that is associated with the levels of the factor (treatments) in a one way ANOVA model. The MODEL statement informs SAS what ANOVA model we are analyzing. Since there is only one CLASS variable listed after the equal sign, SAS knows we are analyzing a one way ANOVA model. Note that at the end of the MODEL statement we have an option of "/P" This generates the printout on page 3 of the SAS output. Usually, we will not ask for the this option as it does not add much information.

The statement OUTPUT OUT=NEW P=PREDY R=RESID; asks SAS to output the predicted values and name them PREDY and also output the raw residuals and name them RESID. The P and R are the SAS names. The PREDY and RESID are the names given by us that we can use in SAS statements.

The PROC PLOT; statement is initiating plots via our next statement: PLOT STRENGTH*PERCENT; This generates a plot of the raw data. Most often we will not ask for this plot.

The next SAS procedure: PROC UNIVARIATE NORMAL; with the statement: VAR RESID; will generate various statistics for the raw residuals. Most importantly, it will generate the p-value of the W statistics that will help us to assess the normality assumption for the random errors. That is why we had to use the OUTPUT OUT statement and give a name for the raw residuals (in our case RESID).

The SAS output 1 Syfiber is 9 pages long. On page 1, we have a table that summarizes the number of levels for our factor and the values of these levels. It also lists the number of observations in the data set.

On page 2 of the SAS output, The ANOVA table is presented that includes the F test for testing the null hypothesis:

$$H_0 : \tau_1 = \tau_2 = \tau_3 = \tau_4 = \tau_5 = 0, \tag{1}$$

which is equivalent to testing:

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5, \tag{2}$$

in our one way ANOVA model:

$$Y_{ij} = \mu + \tau_i + \varepsilon_{ij},\ i = 1, ..., 5,\ j = 1, ..., 5, \tag{3}$$

where τ_i is the effect on the strength of the ith treatment level (i.e. the cotton blending percentage for this level), based on braking level of the ith synthetic fiber and

$$\mu_i = \mu + \tau_i,\ i = 1, ..., 5$$

and μ_i is the average strength for the ith synthetic fiber.

Based on the SAS output we get that the observed value of the F statistic is 14.76 and it $p - value < .0001$. If the underlying assumption for our ANOVA model are valid, then we reject the null hypothesis in (2) and conclude that the average breaking strength for the five synthetic fibers are not equal to each other. We also see that the R-Square coefficient, that measures the amount of total variability explained by our model is about .76, i.e. 76% of variability is explained by our model.

The validity of the underlying assumptions for our model are assessed via the plots in the Fit Diagnostics panel, presented on page 4 of the SAS output, and the W statistic's p-value, generated by PROC UNIVARIATE, on page 8 of the SAS output. The normal probability plot (Residual vs. Quantile plot), presented in the second row on the left on page 4 of the SAS output, look pretty straight without any visible patterns or curvature. Moreover, the p-value of the W statistic given on page is about .18 > .10. Based on that we conclude that the normality assumption for the random errors is valid. Next, we examine the plot of the Residuals vs. Predicted values given in the first row on the left in the Fit Diagnostics pane. There are no visible patterns and the points are scattered randomly about the 0 line. This confirms the linearity assumption and the constant variance assumption for our ANOVA model. From the plot of the RStudent residuals vs. Predicted values, 2nd plot in the first row in the Fit Diagnostic panel, we see that all the values are larger than -2 and smaller that 2.5. This implies that we do not have any outliers. The fact that all the Cook'D values are less than 1 (actually they are less than .3) implies that we do not have any influential observation. Based on this we can confirm that the F statistic and the conclusions we have drawn are valid.

Note that the conclusion that the average breaking strengths of the five synthetic fibers are different from each other, is only the first step in our data analysis. The next important step is to perform a multiple comparison analysis for the means and decide which of the synthetic fiber(s) are significantly stronger that the other.